The basic trigonometric identity

Pythagoras' theorem, divided through by c^2 — and suddenly it is a statement about angles.

LESSON 24 | 1 × 40 MIN | GEOMETRY 8, ТЕМЫ 20–21 | STAGE 9 · BEYOND

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Derive $\sin^2\alpha + \cos^2\alpha = 1$ from Pythagoras' theorem.
- Find one ratio from another using the identity.
- Use the corollaries $\tan\alpha = \sin\alpha / \cos\alpha$ and $\tan\alpha \cdot \cot\alpha = 1$.
- Choose the right sign — for an acute angle every ratio is positive.

TEXTBOOK

National	Темы 20–21, pp. 49–51
Cambridge	Extension beyond Stage 9
Workbook	–

01 Explanation

The derivation

Start from Pythagoras in the triangle of the last lesson and divide every term by c^2:

$$a^2 + b^2 = c^2$$

$$\frac{a^2}{c^2} + \frac{b^2}{c^2} = 1$$

$$\left(\frac{a}{c}\right)^2 + \left(\frac{b}{c}\right)^2 = 1$$

THE BASIC TRIGONOMETRIC IDENTITY

$$\sin^2\alpha + \cos^2\alpha = 1$$

true for **every** acute angle α.

Notation: $\sin^2\alpha$ means $(\sin\alpha)^2$, never $\sin(\alpha^2)$.

Three corollaries

$$\sin\alpha = \sqrt{1 - \cos^2\alpha} \quad \cdot \quad \cos\alpha = \sqrt{1 - \sin^2\alpha}$$

The positive root is the right one every time, because for an acute angle all four ratios are positive.

$$\tan\alpha = \frac{\sin\alpha}{\cos\alpha} \quad \cdot \quad \cot\alpha = \frac{\cos\alpha}{\sin\alpha} \quad \cdot \quad \tan\alpha \cdot \cot\alpha = 1$$

Proof of the first: $\frac{\sin\alpha}{\cos\alpha} = \frac{a/c}{b/c} = \frac{a}{b} = \tan\alpha$. The c cancels.

Using it

Given one ratio, the identity produces all the others without ever drawing a triangle:

$$\sin\alpha = 0.6 \Longrightarrow \cos^2\alpha = 1 - 0.36 = 0.64 \Longrightarrow \cos\alpha = 0.8$$

and then $\tan\alpha = \frac{0.6}{0.8} = 0.75$.

A QUICK SANITY CHECK

Both $\sin\alpha$ and $\cos\alpha$ must lie strictly between 0 and 1, and $\sin^2\alpha + \cos^2\alpha$ must come to exactly 1. If it does not, go back.

02 Worked examples

EXAMPLE 1

Given $\sin\alpha = \dfrac{3}{5}$, find $\cos\alpha$ and $\tan\alpha$.

1. $\cos^2\alpha = 1 - \left(\dfrac{3}{5}\right)^2 = 1 - \dfrac{9}{25} = \dfrac{16}{25}$

   The identity.

2. $\cos\alpha = \dfrac{4}{5}$

   Positive root — the angle is acute.

3. $\tan\alpha = \dfrac{\sin\alpha}{\cos\alpha} = \dfrac{3/5}{4/5} = \dfrac{3}{4}$

ANSWER

$\cos\alpha = 0.8,\ \tan\alpha = 0.75$

EXAMPLE 2 **Given $\cos\alpha = \dfrac{5}{13}$, find $\sin\alpha$ and $\cot\alpha$.**

1. $\sin^2\alpha = 1 - \dfrac{25}{169} = \dfrac{144}{169}$

2. $\sin\alpha = \dfrac{12}{13}$

3. $\cot\alpha = \dfrac{\cos\alpha}{\sin\alpha} = \dfrac{5}{12}$

ANSWER

$\sin\alpha = \dfrac{12}{13}$, $\cot\alpha = \dfrac{5}{12}$

EXAMPLE 3 **Simplify $(1 - \sin\alpha)(1 + \sin\alpha)$**

1. $= 1 - \sin^2\alpha$

   Difference of two squares.

2. $\sin^2\alpha + \cos^2\alpha = 1$

   so $1 - \sin^2\alpha = \cos^2\alpha$

3. $= \cos^2\alpha$

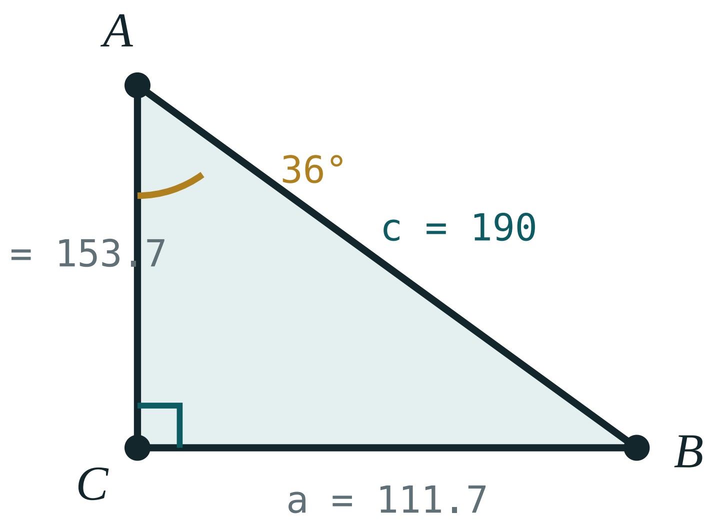
ANSWER

$\cos^2\alpha$

03 Interactive model

Move the angle and watch $\sin^2 + \cos^2$ stay at 1.00 for every position.

sin, cos and the identity



α **36°**	β **54°**	a (opposite) **111.7**	b (adjacent) **153.7**	c (hyp) **190**	sin α **0.59**
cos α **0.81**	tan α **0.73**	$a^2 + b^2$ **36100**	c^2 **36100**		

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Identity	Ayniyat	Тождество
Basic trigonometric identity	Asosiy trigonometrik ayniyat	Основное тригонометрическое тождество
Corollary	Natija	Следствие
Square of the sine	Sinusning kvadrati	Квадрат синуса
Derive	Keltirib chiqarish	Вывести
Divide through	Hadma-had bo‘lish	Разделить почленно
Positive	Musbat	Положительный
Substitute	O‘rniga qo‘yish	Подставить

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$sin^2\alpha + cos^2\alpha$ equals:

- 0
- 1
- 2
- $tan\ \alpha$

If $sin\ \alpha = 0.6$ then $cos\ \alpha$ is:

- 0.4
- 0.8
- 0.36
- 1.4

$tan\ \alpha$ equals:

- $\frac{\cos\alpha}{\sin\alpha}$
- $\frac{\sin\alpha}{\cos\alpha}$
- $sin\ \alpha \cdot cos\ \alpha$
- $1 - cos\ \alpha$

$1 - \sin^2\alpha$ equals:

- $\cos^2\alpha$
- $\tan^2\alpha$
- $\sin\alpha$
- 0

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $\sin\alpha = 0.6$. Find $\cos\alpha$.

   ANSWER 0.8

2. $\cos\alpha = 0.8$. Find $\sin\alpha$.

   ANSWER 0.6

3. $\sin\alpha = \frac{3}{5}$. Find $\cos\alpha$.

   ANSWER $\frac{4}{5}$

4. $\cos\alpha = \frac{12}{13}$. Find $\sin\alpha$.

   ANSWER $\frac{5}{13}$

5. Simplify $1 - \cos^2\alpha$

   ANSWER $\sin^2\alpha$

6. Simplify $1 - \sin^2\alpha$

   ANSWER $\cos^2\alpha$

7. $\tan\alpha \cdot \cot\alpha = ?$

   ANSWER 1

Medium · the standard question

7 PROBLEMS

1. $\sin\alpha = \frac{3}{5}$. Find $\cos\alpha$ and $\tan\alpha$.

ANSWER $\frac{4}{5}, \frac{3}{4}$

2. $\cos\alpha = \frac{5}{13}$. Find $\sin\alpha$ and $\cot\alpha$.

ANSWER $\frac{12}{13}, \frac{5}{12}$

3. $\sin\alpha = \frac{8}{17}$. Find $\cos\alpha$.

ANSWER $\frac{15}{17}$

4. Simplify $(1 - \sin\alpha)(1 + \sin\alpha)$

ANSWER $\cos^2\alpha$

5. Simplify $\sin^2\alpha - 1$

ANSWER $-\cos^2\alpha$

6. $\tan\alpha = \frac{3}{4}$. Find $\cot\alpha$.

ANSWER $\frac{4}{3}$

7. Simplify $\frac{\sin\alpha}{\cos\alpha} \cdot \cot\alpha$

ANSWER 1

Hard · stretch and reason

7 PROBLEMS

1. $\sin\alpha = \frac{7}{25}$. *Find* $\cos\alpha$, $\tan\alpha$ *and* $\cot\alpha$.

ANSWER $\frac{24}{25}, \frac{7}{24}, \frac{24}{7}$

2. *Simplify* $\sin^2\alpha + \cos^2\alpha + \tan\alpha \cdot \cot\alpha$

ANSWER $1 + 1 = 2$

3. *Simplify* $(\sin\alpha + \cos\alpha)^2$

ANSWER $1 + 2\sin\alpha\cos\alpha$

4. *Simplify* $(\sin\alpha + \cos\alpha)^2 + (\sin\alpha - \cos\alpha)^2$

ANSWER 2

5. *Prove that* $\frac{1}{\cos^2\alpha} = 1 + \tan^2\alpha$

ANSWER Divide the basic identity by $\cos^2\alpha$.

6. $\tan\alpha = 2$. *Find* $\sin\alpha$ *and* $\cos\alpha$.

ANSWER $\cos^2\alpha = \frac{1}{5}$, so $\cos\alpha = \frac{1}{\sqrt{5}}$ and $\sin\alpha = \frac{2}{\sqrt{5}}$

7. *Simplify* $\cos^2\alpha - \sin^2\alpha + 2\sin^2\alpha$

ANSWER 1

07 Homework

Homework — 5 problems

Geometry 8, Темы 20–21, pp. 49–51.

1. $\sin\alpha = \frac{15}{17}$. *Find* $\cos\alpha$ *and* $\tan\alpha$.

2. $\cos\alpha = 0.28$. *Find* $\sin\alpha$.
3. *Simplify* $(1 - \cos\alpha)(1 + \cos\alpha)$
4. *Simplify* $\sin^2\alpha \cdot \cot\alpha \cdot \tan\alpha$
5. *Derive* $\sin^2\alpha + \cos^2\alpha = 1$ *from Pythagoras' theorem, with a labelled figure.*